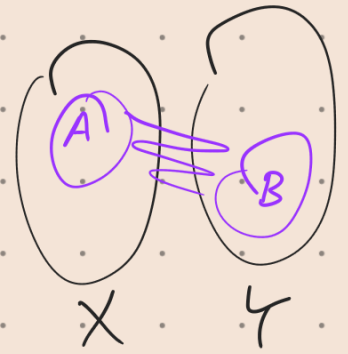


Lect 6 Key Lemmas related to SzRL

• (X, Y) is ϵ -regular if

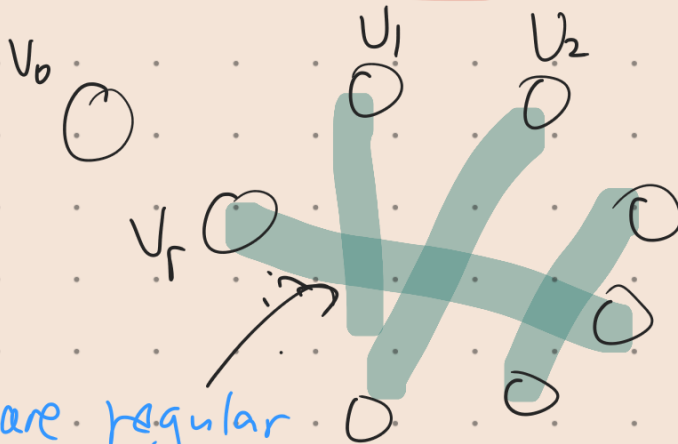
(random like) $\forall |A| \geq \epsilon |X|$
 $\forall |B| \geq \epsilon |Y|$



$$\Rightarrow d(A, B) = \frac{e(A, B)}{|A||B|} = d(X, Y) \pm \epsilon$$

$d(X, Y) = \frac{e(X, Y)}{|X||Y|}$ = prob. of a uniform random X - Y -pair being adjacent

Sz.R.L. $\forall G \exists$ regular partition



$$r = O\left(\frac{1}{\epsilon, m}\right)$$

$$|V_i| = |V_j|$$

Today

- 1) reduced graph (fixed size associated to a reg part. of G)
- 2) embedding lemma
- 3) Counting lemma
- 4) Δ removal lemma

Reduced graph

Def. Given ϵ -regular partition

$$V(G) = V_0 \cup V_1 \cup V_2 \cup \dots \cup V_r \quad \text{of } G$$

given $\delta > 0$ (often $\delta > 10\epsilon$) $R(\epsilon, \delta, V)$

the reduced/cluster graph $R = R(\epsilon, \delta)$

$$\begin{cases} V(R) = [r] \end{cases}$$

$$\begin{cases} ij \in E(R) \text{ iff } \begin{cases} (V_i, V_j) \text{ is } \epsilon\text{-reg.} \\ d(V_i, V_j) \geq \delta \end{cases} \end{cases}$$

Remark Useful also to think of R as

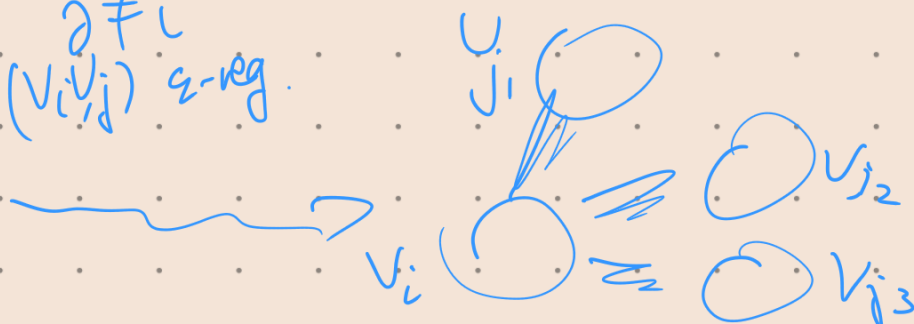
a weighted graph

$$\begin{cases} V(R) = [r] \\ ij \in E(R) \end{cases}$$

$$\begin{cases} (V_i, V_j) \text{ is } \epsilon\text{-reg.} \\ w(ij) = d(V_i, V_j) \end{cases}$$

$$d_R(i) = \sum_{j \neq i} d(V_i, V_j)$$

(V_i, V_j) ϵ -reg.



Ex 1.2 Normalised min-deg is inherited by R

$$\frac{\delta(R) + 1}{r} \geq \frac{\delta(G)}{n} - \underbrace{\delta - \varepsilon}_{\left(= \frac{\delta(G)}{n} - o(1) \right)}$$

Ex 1.3 the edge-density is also retained

$$\frac{e(G)}{\binom{|G|}{2}} \leq \frac{e(R)}{\binom{|R|}{2}} + \underbrace{o(1)}$$

• Property $\left[\begin{array}{l} G \text{ is } H\text{-free} \\ \Leftrightarrow R \text{ has no homo. image of } H \end{array} \right]$



Embedding lemma

• Given a graph F , its blow-up

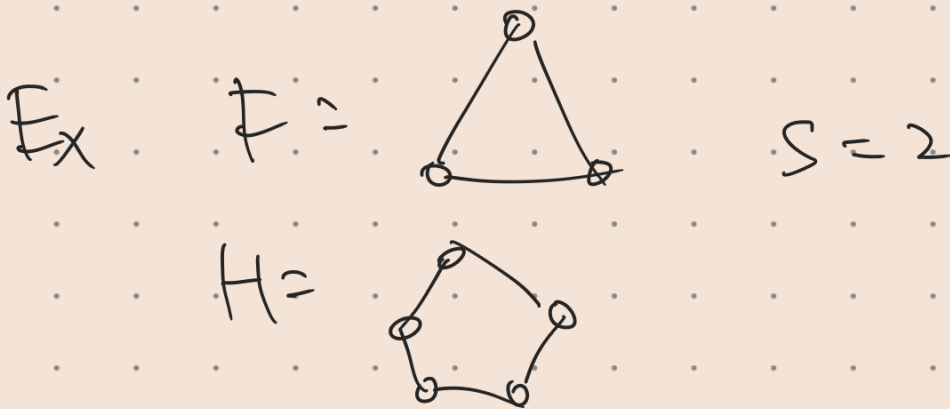
$F(s)$



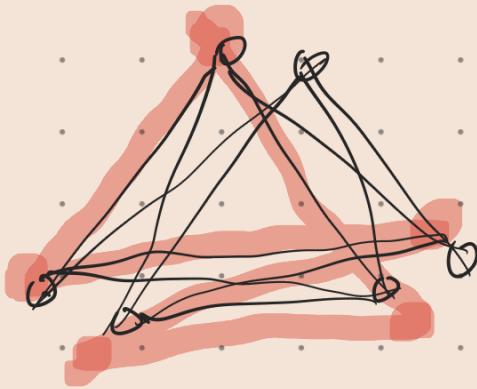
$ux \rightsquigarrow \text{indep set size } S$
 $\text{edge} \rightsquigarrow K_{S,S}$

Observation (*)

$F(S) \supseteq H \iff F \text{ contains a homomorphic image of } H.$



$F(2) =$



Lem 2.1 (Embedding lem) For any $d \in (0,1]$,

$\Delta \geq 1, \exists \epsilon_0 > 0$ s.t. $\nexists F, H. \forall \epsilon \leq \epsilon_0.$

Let G, H graphs with $\Delta(H) \leq \Delta, S \in \mathbb{N}$

and $R = R(\epsilon, d)$ be a reduced graph of G .

Suppose the corresponding reg. parts $V_1, v, \dots, v, V_r$

$|V| = \ell \geq \frac{2s}{d^\Delta}$ Then

$$H \in R(s) \implies H \in G$$

(R contains a homomorphic image of H)

Rank: $|G| = \Theta(\ell)$, $|H|$ could be as large as s . & $s = \Omega(\ell)$.

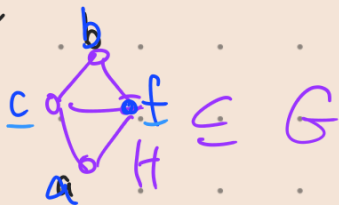
Pf (Special case that $H =$ )

In R , $\exists$ a triangle



$H \in R(2)$

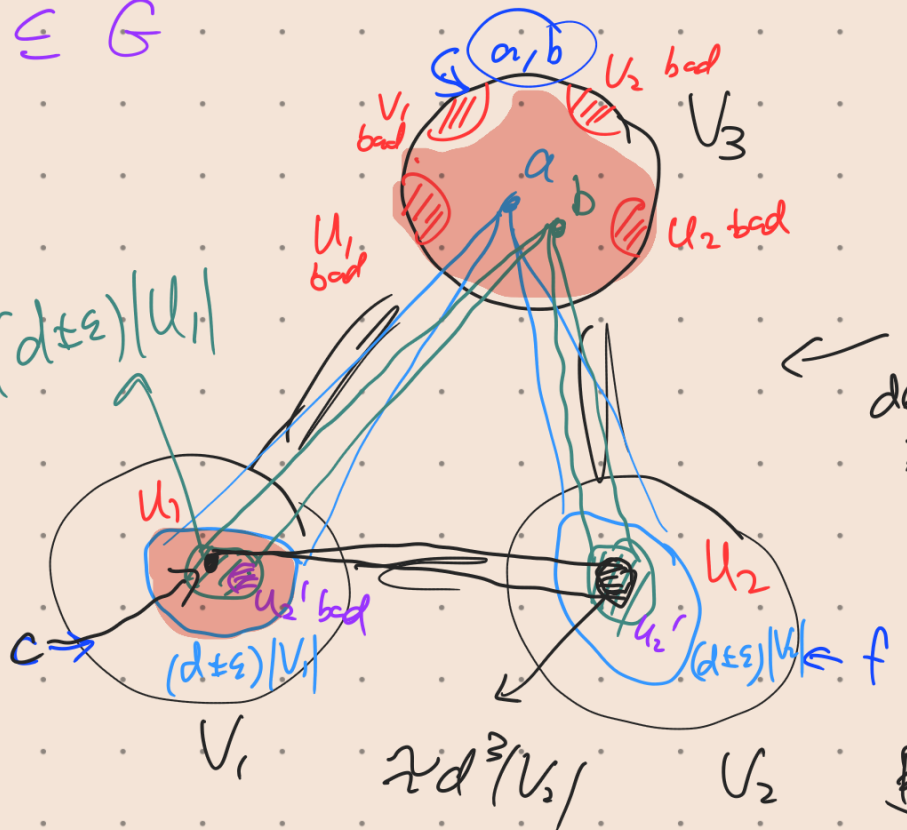
NTS



$V_i, V_j, ij \in \binom{[3]}{2}$

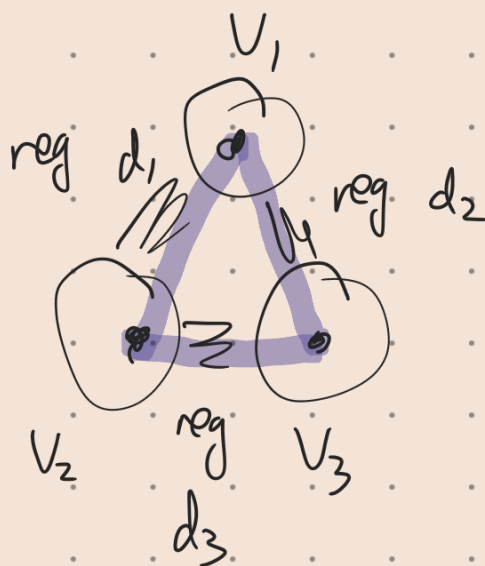
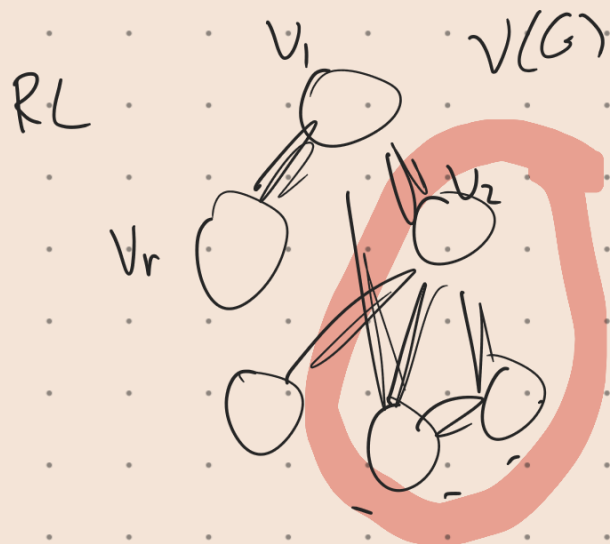
ϵ -reg & $d_{ij} \geq d$

$$d^2/V_i \approx (d \pm \epsilon)/|U_i|$$



Counting lemma

- State here Δ counting lemma (see pdf for H -counting)



if completely random, $\Rightarrow \mathbb{E}(\# \Delta_S) = \binom{|V_1||V_2||V_3|}{d_1 d_2 d_3}$.

Lem 3.1
(Counting len for Δ_S)

If V_1, V_2, V_3 with (V_i, V_j) ε -reg, $\forall ij \in \binom{[3]}{2}$
with $d(V_i, V_j) = d_{ij} \gg \varepsilon$.

$$\Rightarrow \# \Delta_S(V_1, V_2, V_3) = (1 \pm o(\varepsilon)) |V_1| |V_2| |V_3| \underline{d_{12} d_{13} d_{23}}$$

Ex 3.2 Prove counting lemma when $H = K_3$.

Ex 3.3 Formulate & prove counting lemma
for induced copies of $H = C_4$.

Ruzsa - Szemerédi Δ removal lemma

Informally if G is almost Δ -free ($\#\Delta_s = o(n^3)$)

then it can be made Δ -free by removing a negligible amount of edges ($o(n^2)$)

Lem 4.1 (Δ Removal Lem) Given $c > 0$, $\exists a = a(c) > 0$
s.t. T.F.H. for suff. large n .

Let G be n -v graph with $\leq an^3 \Delta_s \Rightarrow$ it can be made Δ -free by removing $\leq cn^2$ edges.

Contrapositive if we can not make G Δ -free by removing few edges (many edge-disjoint Δ_s),
 $\Rightarrow$ then G has lots of Δ_s (positive proportion).

Application $(6,3)$ -thm, Roth's thm

Δ -counting lem $\Rightarrow$ Δ -removal lem.

Cleaning G



$$G \longrightarrow G_R \subseteq G$$

remove all edges

not "corresponding" to R.

Given a reduced graph $R = R(\varepsilon, \delta)$ of G

G_R is obtained as follows:

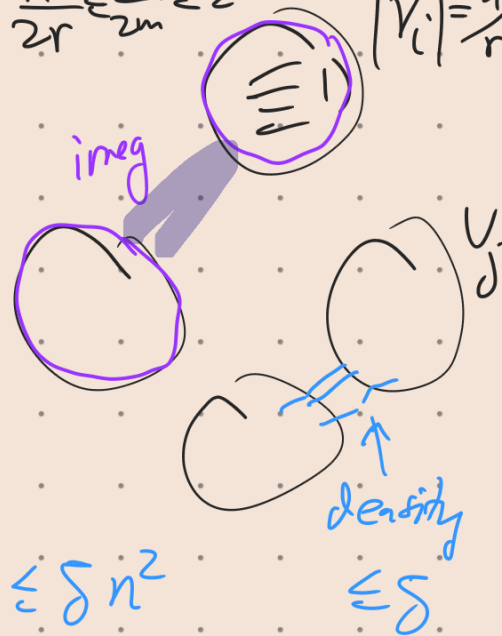
$$e(G_R) = e(G) - o(n^2)$$

1) remove "inner edges": edges in V_i , $i \in [r]$

$$r \geq m \geq \frac{1}{\varepsilon} \Rightarrow \# \text{ inner edges} \leq \binom{n/r}{2} \cdot r \leq \frac{n^2}{2r} \leq \frac{n^2}{2m} \leq \frac{1}{2} \varepsilon n^2 \quad |V_i| = \frac{n}{r}$$

2) edges btw irreg pairs

$$\# \text{ irreg edges} \leq \frac{n^2}{r^2} \cdot \varepsilon r^2 = \varepsilon n^2$$



3) edges btw sparse pairs

$$\# \text{ edges in sparse pairs} \leq \delta \cdot \frac{n^2}{r^2} \binom{r}{2} \leq \delta n^2$$

$$\text{total \# edges deleted} = O((\varepsilon + \delta)n^2) = o(n^2)$$



G_R

$\longrightarrow G_R$

PF of Δ removal lem

$$\left(\forall c \exists a = a(c) \leq an^3 \Delta_s \xrightarrow{\text{remove } cn^2 \text{ edges}} \Delta\text{-free} \right)$$

PF Suppose not. I.e.

$\exists c > 0$, s.t. $\forall a > 0$, $\exists$ a counterexample G
 i.e. $\# \Delta_s \text{ in } G \leq an^3$, but
 removal of any cn^2 edges does not destroy
 all Δ_s .

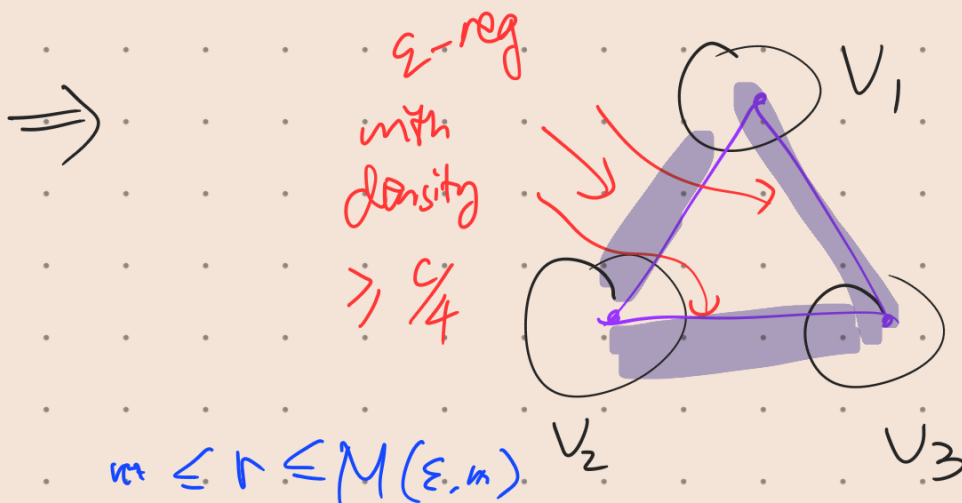
• Apply Sz.R.L. with $\epsilon < c$, $m = \frac{1}{\epsilon}$

$\Rightarrow \epsilon$ -reg part $V_1 \cup \dots \cup V_r$ & $R = R(\epsilon, \frac{c}{4})$

Let $G_R \subseteq G$ be the cleaned graph.

• Now $e(G_R) \geq e(G) - cn^2/2$

assumption on $G \Rightarrow \exists \Delta$ in G_R .



by counting lemma

$$\begin{aligned} \Rightarrow \# \Delta_s \text{ in } G_R &\geq (1 - 100\epsilon) \left(\frac{n}{r} \right)^3 \cdot \left(\frac{c}{4} \right)^3 \\ &\geq \frac{1}{2} \left(\frac{c}{4m} \right)^3 n^3 \\ &\geq \frac{1}{2} \left(\frac{c}{4M} \right)^3 n^3 > an^3 \end{aligned}$$